

ST3MML Assignment 1

This assignment is worth 15% of the overall module mark for ST3MML.

The deadline for handing in your solutions to this assignment is 12.00 (noon) UK time on 11/11/25. This deadline will be strictly adhered to, unless you are granted an extension under the Assessment Adjustments policy or have a DAS-agreed extension as a reasonable adjustment in your Individual Learning Plan. Late work submitted up to 3 calendar days after the original deadline or DAS-agreed extension will be subject to the University penalty of a deduction of 10% of the available marks per calendar day late until the threshold of 40% is reached (a penalty will not be applied to a mark below 40%) – work submitted later than 3 calendar days will receive a mark of 0. Work submitted late after any other (non-DAS-agreed) extension will also receive a mark of 0.

Use of GAIT:

This assignment is a Category 2 assignment with respect to the use of Generative AI tools. This means that GAIT may be used to support student learning and development, but you should not include any GenAI output in your report. Please note the following:

- ***If you choose to use GAIT, you may use either the freely available Microsoft Copilot (<https://copilot.microsoft.com/>) or Google Gemini (<https://gemini.google.com/>) I am allowing only these two freely available tools in order to maintain fair access for all.***
- ***The use of GenAI is not a requirement and the assignment can be completed without it.***
- ***Generative AI tools may not give you correct results, so you should carefully check all output yourself.***
- ***Before deciding whether to use GenAI, please consider the negative environmental effects of using these tools.***
- ***If you do use Microsoft Copilot or Google Gemini then you must state this clearly at the top of your assignment., You must also include the prompts/questions you asked in an appendix – there is no need to include the responses it gave you.***
- ***If you do not follow these instructions, then you may be at risk of being highlighted to the School Director of Teaching and Learning, according to the [Academic Integrity and Academic Misconduct policy](#).***

You may work as an individual or as a pair. If you are working as a pair, you must still submit the work to Blackboard individually. You must choose a name for your pair and this name must be shown clearly at the top of the work you submit. This is essential to avoid any confusion over copying when the assignments are marked.

Multivariate analysis of physiological measurements of elite athletes across five sports.

In this assignment you will carry out the analysis of a multivariate dataset. You should work through the tasks and write up your work as a professional report which could be understood by a fellow undergraduate who has not studied the ST3MML module. **Page limit: a maximum of 8 sides of A4, standard font size (no smaller than size 12).** Any output or text beyond this limit will not be read/marked. You must create one Word document or PDF file of your work which includes all graphs and R output. All of your R graphs must include the name of your pair in the title or as a subtitle.

The data

The dataset used in this assignment is a subset from the Australian Institute of Sport (AIS) biometric data, which includes various physiological (and other) measurements of elite athletes across various sports disciplines. The original AIS dataset has been widely used in sports science research and statistical classification exercises (Cook & Weisberg, 1994).

The version used here focuses on female elite athletes from five selected sports—Gymnastics, Rowing, Swimming, Basketball, and Track (Sprinting). The file *ST3MMLsport.csv* containing the data is available on Blackboard.

Variables:

Var	Description	Units
rcc	Red cell count	millions per cubic mm
wcc	White cell count	thousands per cubic mm
hc	Haematocrit	percentage (%)
hg	Haemoglobin	grams per decilitre (g/dL)
Fe	Plasma ferritin concentration	micrograms per litre (µg/L)
bmi	Body Mass Index	kg/m ²
ssf	Sum of skin folds (7-site method)	millimetres (mm)
pcBfat	Body fat percentage (estimated from ssf)	percentage (%)
lbm	Lean Body Mass	kilograms (kg)
ht	Height	centimetres (cm)
wt	Weight	kilograms (kg)
sex	Biological sex	categorical: "f" or "m"
sport	Athlete's sport/event group	categorical (see above)

Each row corresponds to an individual athlete.

Task 1

Begin by performing relevant exploratory data analysis on the dataset. Then, following the approach taught in class, use a suitable multivariate method to explore the separation of the groups (sports) in multivariate space. Justify your choices and fully interpret your findings. Provide clear plots where appropriate.

[25 marks]

Task 2

You are supplied with the following measurements for two additional athletes whose sport categories are unknown:

rcc	wcc	hc	hg	ferr	bmi	ssf	pcBfat	lbm	ht	wt
4.35	7.8	41.4	14.1	30	22.03	117.8	23.3	48.32	169.1	63
4.53	5	40.7	14	41	17.79	56.8	12.55	38.3	156.9	44.8

Following on from your results in Task 1, obtain scores for two discriminant functions for each of these athletes. Show the position of the new athletes with regard to the five

groups on a suitable plot. Comment on the position and the likely sport each athlete represents.

Choose **one** of the new athletes and perform a formal classification by hand, using the methods taught in class. This should be based on all of the discriminant functions that you have identified as important in Task 1. You should show enough working to make your approach clear.

Note: To carry out this task you will need to mean centre the measurements for the unknown athletes and you will need to obtain scores for group means. This mean centering can easily be done in R, using the approach shown in the PC class. The scores for group means for each discriminant function can be found in R using 'name of your candisc output\$ means,'

[20 marks]

Task 3

In R, obtain an estimate for the Apparent error rate (APER) by randomly splitting the dataset into training set (70%) and test set (30%). Comment critically on this approach and on your findings. *Important: To ensure consistent results across students, please set the random seed using `set.seed(123)` before performing the split.*

[5 marks]

Marks will be awarded for:

Choosing appropriate methods for displaying and analysing the data.

Correctly implementing methods in R, and by hand (when requested).

Reporting and interpreting results correctly and in sufficient detail.